

Recipe Finder Web Application – Documentation

Project Title: Recipe Finder Web Application

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Course: Web Programming Project

1. Introduction

The **Recipe Finder Web Application** is a responsive, multilingual web platform that allows users to search, explore, and save recipes from around the world. It is developed using **React.js**, **React Bootstrap**, and integrates with **TheMealDB API**, a free and public RESTful API that provides recipe and meal information, which I found from the list of API sample provided by the lecturer.

This project demonstrates modern front-end web development principles, including API integration, internationalization (i18n), responsive design, dark mode implementation, and local data persistence using localStorage.

2. Objectives

The main objectives of this project are:

- To design a **user-friendly and visually appealing interface** for recipe search and browsing.
- To implement **multilingual support** using react-i18next (English, Finnish, and Swedish).
- To connect with an **external API (TheMealDB)** to retrieve real recipe data.
- To allow users to **save favorite recipes** for later viewing.
- To provide a **responsive layout** suitable for both desktop and mobile devices.
- To implement **dark mode** and **scroll-to-top functionality** for better user experience.

3. System Overview

The Recipe Finder is a **Single Page Application (SPA)** built with React.js and React Router DOM. The app uses **Vite** as its build tool for faster development and **Bootstrap 5** for responsive design.

The layout is composed of:

- **Navigation Bar:** Includes the logo, search bar, language selector, favorites counter, and dark mode toggle.

- **Home Page:** Displays recipe categories, regions, and random recipe suggestions.
- **Search Page:** Shows recipes based on the user's search query.
- **Favorites Page:** Displays recipes saved locally by the user.
- **Meal Detail Page:** Displays ingredients and instructions for the selected recipe.
- **Scroll-To-Top Button:** A floating button to quickly return to the top of the page.

4. Application Architecture

Technology Stack

- **Frontend Framework:** React.js (with Vite)
- **Styling:** React Bootstrap & Bootstrap 5
- **Routing:** React Router DOM
- **Internationalization:** react-i18next & i18next
- **External API:** TheMealDB
- **State Management:** React Hooks (useState, useEffect)
- **Data Storage:** localStorage for favorites and dark mode preference

Component Overview

1. **App.jsx:** Main application wrapper with routes and shared layout.
2. **AppNavbar.jsx:** Navigation bar including search, language selection, and dark mode toggle.
3. **Home.jsx:** Displays recipe categories, regions, and random meal cards.
4. **Search.jsx:** Displays search results based on API queries.
5. **Favorites.jsx:** Shows user's saved favorite recipes from local storage.
6. **MealDetail.jsx:** Displays detailed recipe instructions and ingredients.
7. **ScrollToTopButton.jsx:** Provides easy navigation back to the top of the page.

5. External API – TheMealDB

The application integrates with **TheMealDB API** to fetch meal data in real-time. The API returns all meal names and instructions **in English only** — this means translations are limited to the user interface.

API Endpoints Used:

- Search meals by name:
<https://www.themealdb.com/api/json/v1/1/search.php?s=chicken>
- Filter by category:
<https://www.themealdb.com/api/json/v1/1/filter.php?c=Seafood>
- Filter by region:
<https://www.themealdb.com/api/json/v1/1/filter.php?a=Canadian>
- Lookup meal by ID:
<https://www.themealdb.com/api/json/v1/1/lookup.php?i=52772>
- Get random meal:
<https://www.themealdb.com/api/json/v1/1/random.php>

All API requests return JSON data, which is displayed dynamically using React components and styled with Bootstrap.

6. Multilingual Support

The app supports **three interface languages**:

- GB English (en)
- FI Finnish (fi)
- SE Swedish (sv)

Note:

Because **TheMealDB API only provides data in English**, only the **interface elements** (buttons, labels, titles, placeholders, etc.) are translated.

Meal names and instructions remain in English.

7. Dark Mode

The dark mode toggle in the navbar lets users switch between **light and dark themes**. This state is saved in localStorage, so the user's preference is retained even after refreshing the page.

8. Scroll-To-Top Button

The floating **Scroll-to-Top button** appears once the user scrolls beyond 400px. It uses smooth scrolling for a better experience and matches the app's yellow accent color.

9. Responsive Design

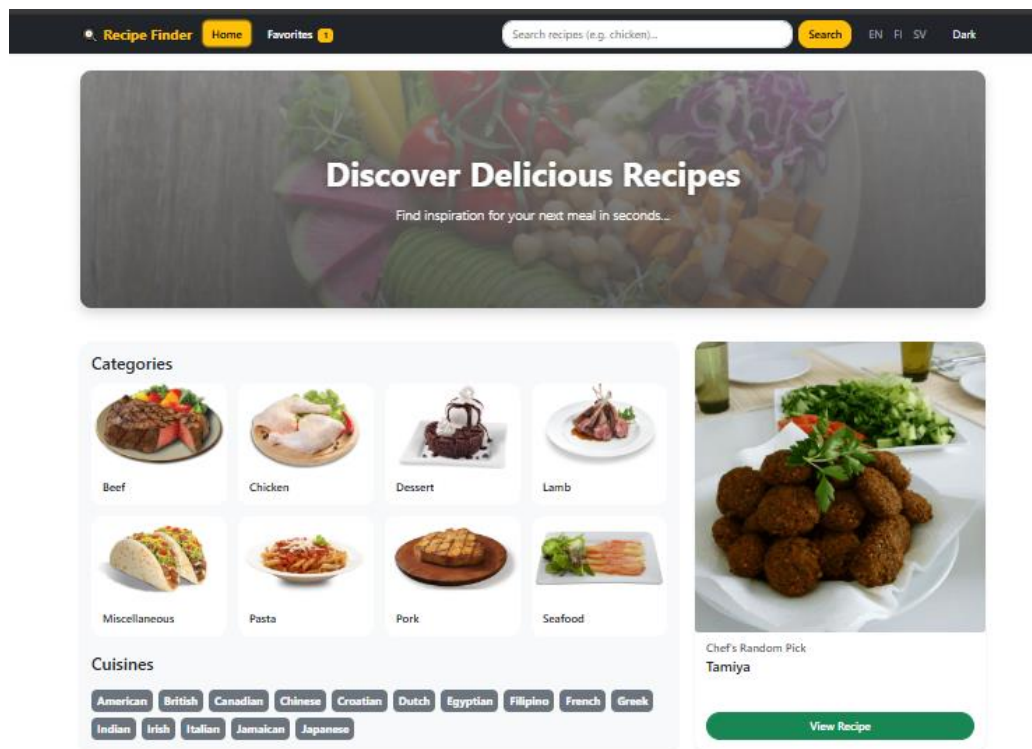
The app uses Bootstrap's grid system to ensure proper display across all screen sizes:

- **Desktop:** 3–4 recipe cards per row.
- **Tablet:** 2 cards per row.
- **Mobile:** 1 card per row.

All cards, text, and buttons scale dynamically.

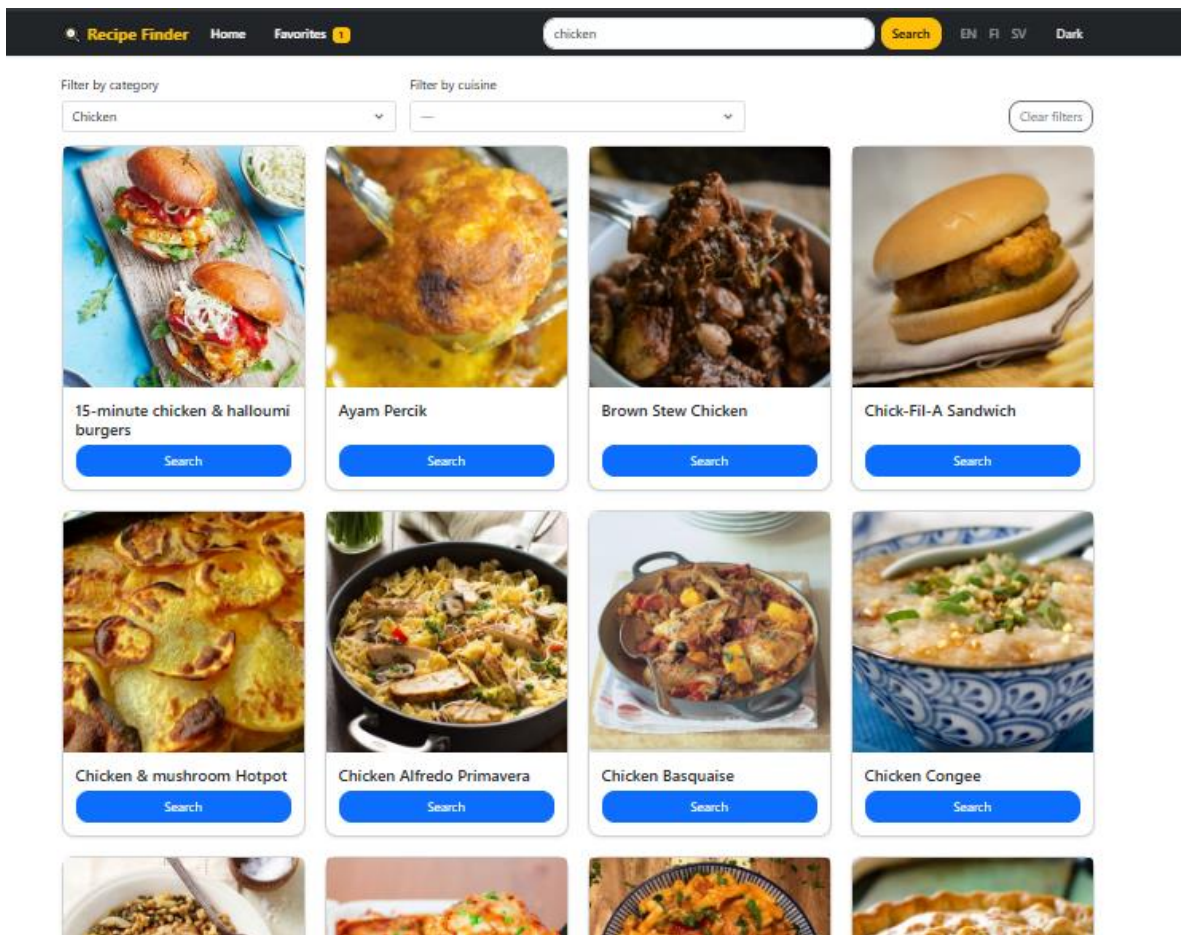
10. Application Pages and Screenshots

1. **Home Page** – Displays hero banner, search bar, recipe categories, and random recipes.
Home Page (Light Mode)



2. **Search Page** – Displays real-time recipe results fetched from the API.

You can also apply a filter to your search



3. Meal Detail Page – Shows full recipe instructions, ingredients, and image. *Meal Detail Page*

Recipe FinderHomeFavorites 1

chickenSearchENFI SVDark



Add to Favorites

ChickenMexican

Chicken Enchilada Casserole

Ingredients:

- Enchilada sauce 14 oz jar
- shredded Monterey Jack cheese 3 Cups
- corn tortillas 6
- chicken breasts 2

Instructions:

Cut each chicken breast in about 3 pieces, so that it cooks faster and put it in a small pot. Pour Enchilada sauce over it and cook covered on low to medium heat until chicken is cooked through, about 20 minutes. No water is needed, the chicken will cook in the Enchilada sauce. Make sure you stir occasionally so that it doesn't stick to the bottom.

Remove chicken from the pot and shred with two forks.

Preheat oven to 375 F degrees.

Start layering the casserole. Start with about ¼ cup of the leftover Enchilada sauce over the bottom of a baking dish. I used a longer baking dish, so that I can put 2 corn tortillas across. Place 2 tortillas on the bottom, top with ½ of the chicken and ½ of the remaining sauce. Sprinkle with ½ of the cheese and repeat starting with 2 more tortillas, then chicken, sauce, cheese. Repeat with last layer with the remaining ingredients, tortillas, chicken, sauce and cheese.

Bake for 20 to 30 minutes uncovered, until bubbly and cheese has melted and started to brown on top. Serve warm.

Note: Recipe descriptions are available only in English.

4. Favorites Page – Displays all saved recipes stored in local Storage.

Recipe FinderHomeFavorites 3

Search recipes (e.g. chicken)SearchENFI SV



Chicken Karaage

Search



Apple & Blackberry Crumble

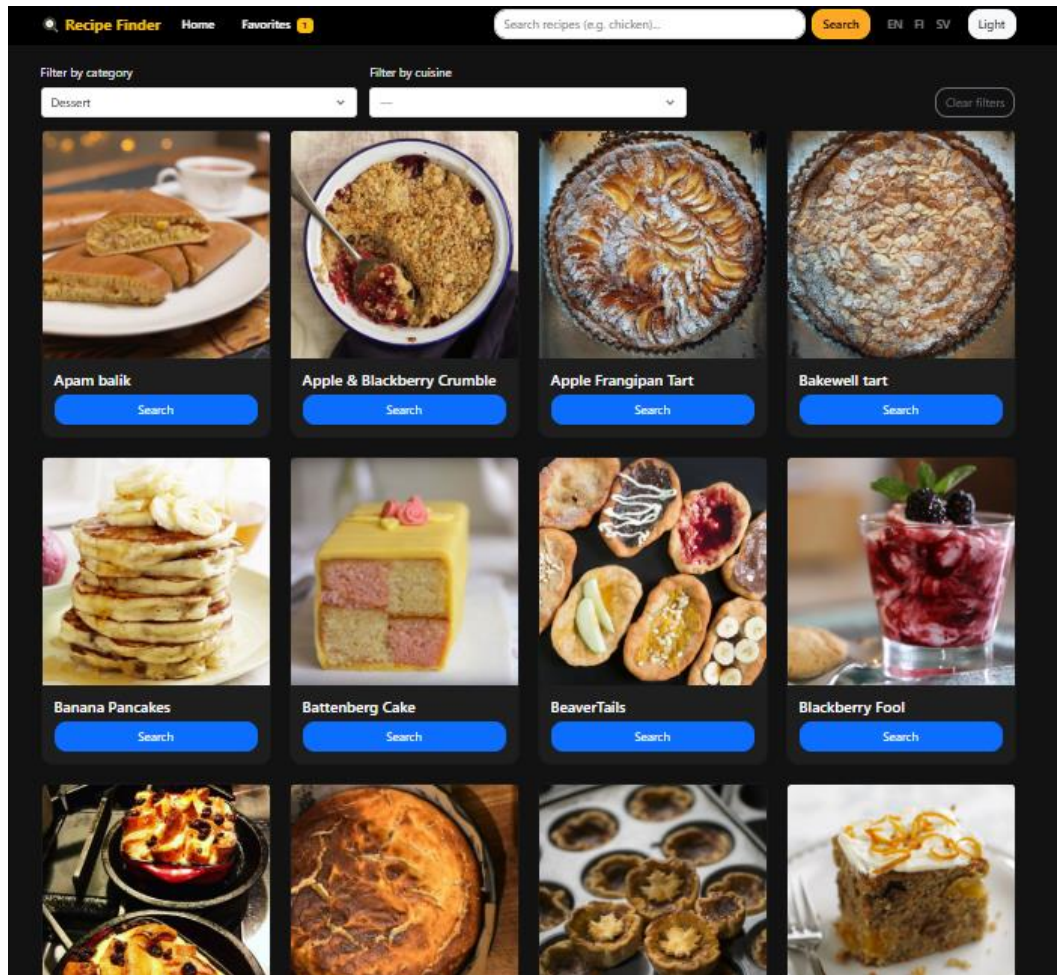
Search



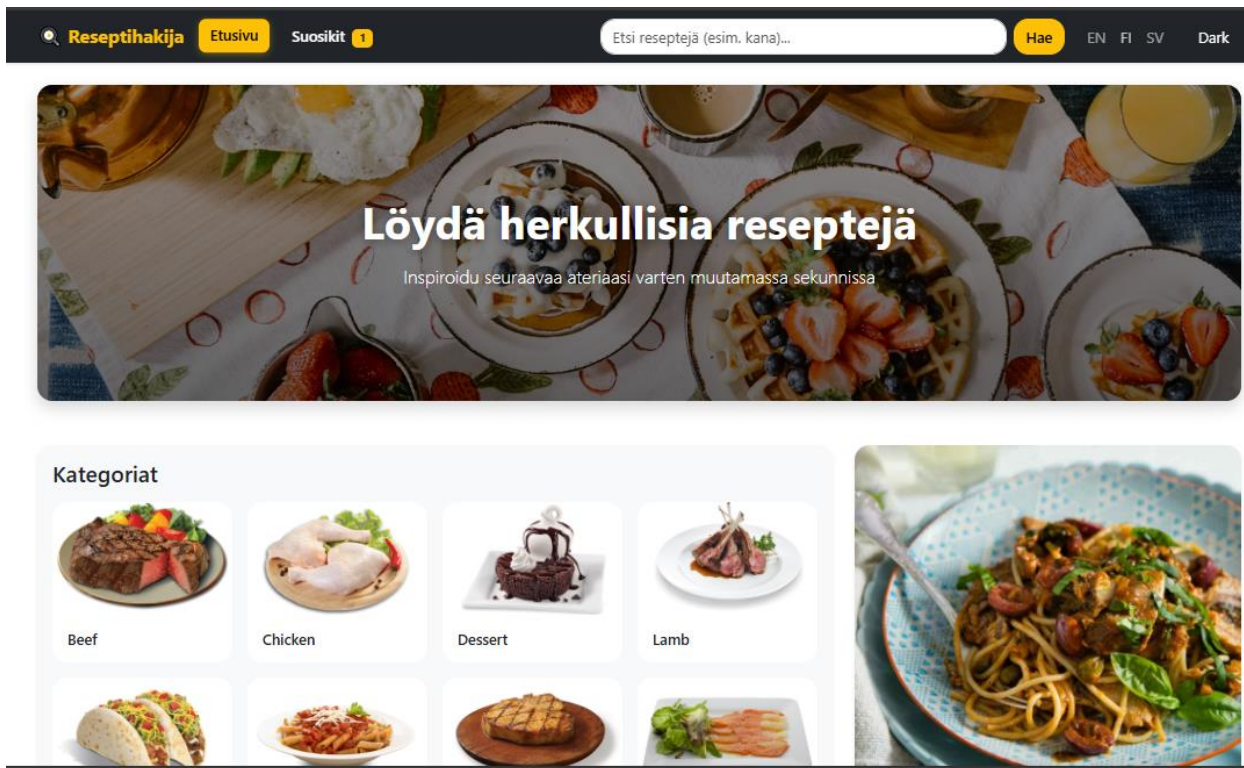
Cajun spiced fish tacos

Search

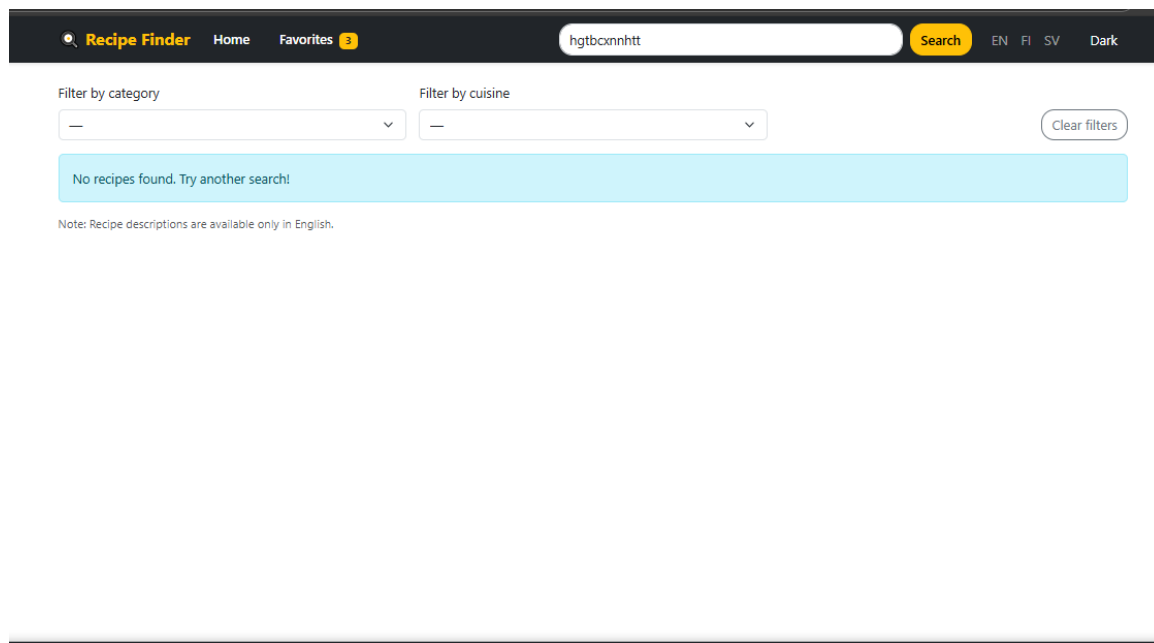
5. **Dark Mode View** – Shows interface in dark mode.



6. Language Translation View – Demonstrates Finnish or Swedish translation of UI.



7. No Result found View – When there is a wrong search



11. Installation and Setup

How to Run the Application

Extract the ZIP file

- After downloading the project ZIP file, extract it to a folder

Open in VS Code

- Open the extracted folder in Visual Studio Code or any preferred code editor.

Install dependencies

- Make sure you have Node.js and npm installed. In the terminal, run: `npm install`,
- `npm run dev`
- VITE v5.x.x ready in 300ms
- Local: `http://localhost:5173/`

Open your browser and go to eg **`**http://localhost:5173/**`** to view the app.

12. How to Use the Application

1. Type a keyword (e.g., *chicken*) in the search bar and press “Search.”
2. Browse recipe cards and click one to view details.
3. Click the Add to favorite button to add a recipe to your **Favorites**.
4. Use **EN / FI / SV** buttons in the navbar to switch the interface language.
5. Use the button to toggle **Dark Mode**.
6. Explore **Random Recipes** on the home page each time you visit.

13. Data Flow and Storage

- **API Data:** Fetched asynchronously from TheMealDB using Axios or `fetch()`.
- **Local Storage:** Stores favorites and dark mode preferences.
- **State Management:** Controlled using React Hooks (`useState`, `useEffect`).
- **Routing:** Managed by React Router without page reloads.

14. Testing and Validation

The application was tested on:

- Google Chrome
- Microsoft Edge

- Mozilla Firefox

Tested Scenarios:

- Searching for recipes
- Switching between languages
- Adding/removing favorites
- Toggling dark mode
- Testing mobile responsiveness

All functionalities performed as expected.

15. Limitations

- The **TheMealDB API only supports English**, so translations apply only to interface text (buttons, titles, placeholders).
- Recipe names, descriptions, and instructions remain in English due to API limitations.
- The API's dataset is limited to meals already available on TheMealDB platform.

Conclusion

The **Recipe Finder Web Application** successfully meets all objectives of the project. It demonstrates integration with a public API, responsive interface design, multilingual support, and modern front-end development practices using React.

The app provides a clean, interactive, and educational experience for users to explore global recipes.

Future improvements could include user accounts, custom APIs, or support for multiple recipe data sources.